Data Creators › Day 3 — run sheet

Day 3 run sheet — dbt practical and consolidation

Instructor-only companion to the anchor notebook, in the notebook's order: the same numbered sections and labelled parts, each with its time, purpose, what is on screen, what to say, what to ask at the whiteboard (with the answers rooms give and what to add), the demo steps with their coaching, fallbacks, the transition and what to do if ahead or behind. Keep the notebook on the projector and this beside it. Times follow the instructor plan; adapt them to the room and keep the sequence.

Before the day

- Section 1 builds a small dbt project from two CSVs in `demos/price_sheet/`, one cell at a time; its first cell resets the folder. Run the section once before the room arrives.
- The walkthrough cells run the same project as the learners' challenge, on DuckDB in `demos/portfolio_analytics/`. Run `bash demos/portfolio_analytics/demo.sh stretch` once before the day; the cells then rebuild pieces on demand.

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Sections

#	Section	Time	What happens
1	dbt from zero: one transformation through the whole life cycle	09:00–09:50	Recap: yesterday's platform in four pictures · Before we start: two things you could not judge yesterday · Demo: project, profile, seeds, source, model, Jinja, ref, tests, build, docs
2	The dbt practical: launch, Phase 1 and the walkthrough	09:50–13:00	Launch: the pipeline · Challenge 9: dbt practical · Walkthrough: Phase 1 on DuckDB · Break · Walkthrough: Phase 2 on DuckDB · Circulating: Phase 2 to lunch · Lunch
3	Advanced dbt, then complete and extend the practical	13:00–14:20	Demo: a loop, a macro, a custom test, snapshot, incremental, unit test · Challenge 9 continued: Phase 2 and the stretch · Break
4	Consolidation	14:20–15:00	Activity 9b: second look at yesterday's AI-generated dbt · Debrief: what did we learn? · Activity 9: test review in pairs · Close

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Timeline

§	Time	Block	Type	On screen
1	09:00–09:10	Recap: yesterday’s platform in four pictures	Recap	Anchor › section 1, Recap (four Day 2 diagrams)
1	09:10–09:50	dbt from zero: one transformation through the whole life cycle	Demo	Anchor › section 1, top to bottom (demos/price_sheet)
2	09:50–10:00	Challenge 9 — dbt practical: launch and first sections	Practical	Anchor › Launch box; learners in challenges/v1/02-dbt-pipeline
2	10:00–10:15	Break	Break	
2	10:15–11:00	Continue dbt practical	Practical	Anchor › Walkthrough cells (for support)
2	11:00–12:00	Continue dbt practical: Phase 2 to lunch	Practical	Anchor › Circulation checklist and symptom table; learners in challenges/v1/02-dbt-pipeline
2	12:00–13:00	Lunch	Break	
3	13:00–13:25	Advanced dbt: a loop, a macro, a custom test, snapshot, incremental, unit test	Demo	Anchor › section 3 demo cells (demos/price_sheet, then demos/portfolio_analytics)

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§	Time	Block	Type	On screen
3	13:25–14:05	Complete and extend dbt practical	Practical	Anchor › Launch box and walkthrough cells; learners in challenges/v1/02-dbt-pipeline
3	14:05–14:20	Break	Break	
4	14:20–14:35	Second look at yesterday's AI-generated dbt	Pair activity	Anchor › Activity 9b card; learners in the project Copilot and Catalyst generated on Day 2
4	14:35–14:45	Consolidation	Debrief	Anchor › Recap table
4	14:45–14:55	Test review in pairs	Pair activity	Anchor › Activity 9 card
4	14:55–15:00	Closing discussion	Debrief	Anchor › Before learners leave

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1. dbt from zero: one transformation through the whole life cycle

Recap — yesterday's platform in four pictures · 09:00–09:10

Purpose. Put everyone on the same page before dbt: where the tables live, what runs the SQL, and which box of the tools picture today zooms into.

On screen. Anchor › section 1, Recap (four Day 2 diagrams)

Instructor notes.

- Ten minutes, four pictures from Day 2, two or three lines each. Ask one question per picture and let the room answer; do not re-teach.
- End on the tools picture with a finger on the dbt box: today is that box, tomorrow is the Airflow box above it.
- Anyone who missed Day 2: point them at the Day 2 anchor's section 3 for the full version.

Say.

- "One line each: what is a lakehouse, what does Databricks do with your SQL, what is a Delta table?"
- "dbt holds no data and does no computing. It sends SQL; everything on these pictures is what receives it."

Transition. So what is in a dbt project? Two CSVs and an empty folder.

If ahead or behind. Short of time: the under-the-hood picture and the tools picture only.

Demo — one price sheet, from two raw files to a tested, documented table · 09:10–09:50

Purpose. Show every kind of file in a dbt project being written and run once, on a small example, so the challenge's tasks are recognisable: project and profile, seeds, a source, a model, Jinja, ref, materialisation, tests failing and fixed, build, docs.

On screen. Anchor › section 1, top to bottom (demos/price_sheet)

Instructor notes.

- Present from the notebook; no whiteboard. Each cell writes one file (%%writefile) or runs one command. Read the file, ask the room to predict the output, run it.

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- About five minutes a step, forty minutes in all. Steps 5 and 6 (two tests fail on the HSBC rows, one line fixes it, the build goes green) are the point: never skip them.
- At Step 3, spend two minutes on the Jinja table: learners meet source(), ref() and config() in the first hour of the challenge. Loops and macros wait until 13:00.
- Run the section once before the room arrives so dbt's first-run delay is gone; the first cell resets the folder each time.

Say.

- "You write a SELECT. dbt writes the CREATE, works out the order, and checks the result."
- "The run was green and three prices were missing. A successful run says the SQL executed, not that the answer is right."
- "A test is only a query for the rows that break a rule. No rows, it passes."

Demo — dbt from zero, eight steps. A project that starts as two CSVs in `demos/price_sheet/seeds/`; each cell writes one file or runs one command.

1. Read each file aloud before running the command that uses it; ask the room to predict the output.
2. Steps 5 and 6 are the point of the demo: two tests fail on real rows, one line fixes the model, the build goes green. Do not skip them.
3. Step 3's compiled SQL is the Jinja explanation: curly braces in, plain SQL out. Learners meet source(), ref() and config() within the first hour of the challenge.
4. `demos/price_sheet/README.md` lists every step, file and expected result; `bash demos/price_sheet/reset.sh` starts again.

Transition. The same kinds of file, more of them: the Portfolio Analytics challenge.

If ahead or behind. Behind at 09:40: skip Step 7 (the singular test) and say it is the pattern of C6. Ahead: open the docs site and click through the lineage graph.

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2. The dbt practical: launch, Phase 1 and the walkthrough

Launch — the pipeline you will build today · 09:50–10:00

Purpose. Launch the assigned dbt project, explain the task sequence and the checks, and get everyone into Phase 1 (sources and staging) before the break.

On screen. Anchor › Launch box; learners in challenges/v1/02-dbt-pipeline

Instructor notes.

- Five-minute launch from the anchor: where to work, the first three commands, what a green section looks like, what reset deletes.
- Then circulate with the three-point checklist. Collect questions in the Q&A table for the recap.
- Break at 10:00; most will be in S1 or S2.

Say.

- “The README teaches dbt as you go: concept, task, hint, check, a variation, then the solution. Attempt before you open the solution.”

Challenge 9 — dbt practical: Phase 1 first · 09:50–10:00

Challenge — dbt practical (challenge 02). `challenges/v1/02-dbt-pipeline/README.md` · about 3 hours across the day.

Say.

- “Attempt each section before you open its solution. The solution is for comparing, not for copying.”
- “You can run the raw dbt commands from the project folder. If dbt says it cannot find your credentials, open a fresh terminal so mise loads the environment, or rerun mise run setup:databricks.”

Walkthrough — Phase 1 on DuckDB: sources and staging · 09:50–10:00

Demo — the same project, on DuckDB, section by section. The learners’ challenge 02 solution ported to DuckDB in `demos/portfolio_analytics/`: one cell per README section, for “what should I see?”.

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1. Same model names, layers, tags and tests; the Databricks-only post-hook is a no-op macro here.
2. Use a cell when someone asks what they should see; `demos/portfolio_analytics/README.md` maps each cell to its README section.
3. Say once: the SQL is the same; only the target differs.

Transition. Break; carry on after it.

If ahead or behind. Anyone who knows dbt: skip to Phase 2 and the stretch list.

> ☕ Break · 15 min · back at 10:15

Walkthrough — Phase 2 on DuckDB: intermediate, marts, tests, docs · 10:15–11:00

Purpose. Phase 1 complete for most; Phase 2 started by some; blockers resolved; questions collected.

On screen. Anchor › Walkthrough cells (for support)

Instructor notes.

- Use the anchor's DuckDB walkthrough cells to show an expected result when someone is stuck; never type in their project.
- Access problems to the platform contact; code problems to the pair next to them first.

Say.

- "Could this join pass its tests and still double the reported value?"

Transition. At 11:00 the practical carries on to lunch.

Circulating — Phase 2 to lunch: three questions per learner, and the symptom table · 11:00–12:00

Purpose. Phase 1 complete for everyone; Phase 2 under way for most; each learner asked the three circulation questions once before lunch.

On screen. Anchor › Circulation checklist and symptom table; learners in challenges/v1/02-dbt-pipeline

Instructor notes.

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- No platform session this week: its content (Airflow at CG, Unity Catalog and the three catalog types, contracts, the vended repository) was covered on Day 2, so the hour goes to the practical.
- Walk the room with the three-point checklist; use the symptom table for the stalls; keep adding to the Q&A table for the recap.
- Platform questions that come up: answer from Day 2's platform section, or log them in the Q&A table for the platform team.

Say.

- "Show me the grain of this model, the key on the right-hand side of the join, and what happens to a row with no match."

Transition. Lunch.

If ahead or behind. Anyone still in Phase 1 at 11:30: pair them with a neighbour who has finished S7.

> 🍽 Lunch · 60 min · back at 13:00

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3. Advanced dbt, then complete and extend the practical

Demo — Jinja that writes SQL: a loop, a macro, then the production pieces · 13:00–13:25

Purpose. Show what Jinja does beyond filling in names, and the pieces a production project adds, so learners recognise them in the vended project and in tomorrow's bitemporal macro.

On screen. Anchor › section 3 demo cells (demos/price_sheet, then demos/portfolio_analytics)

Instructor notes.

- Two cells on this morning's price sheet (a for loop, a macro), then four on the Portfolio Analytics copy (dispatching macro and generic test, snapshot, incremental, unit test).
- After each cell show the compiled SQL: the lesson is that Jinja runs at compile time and the warehouse sees plain SQL.
- None of this is needed to finish the challenge; say so, so nobody abandons Phase 2 for it.

Say.

- "A macro is a function that returns SQL text. If you have pasted the same expression into three models, it wants to be a macro."
- "A snapshot keeps the history of rows that change; an incremental model avoids rebuilding rows that do not."

Transition. Back to the practical: Phase 2, then your choice of stretch.

If ahead or behind. Short of time: the loop, the macro and the snapshot; point at the other cells.

Challenge 9 continued — Phase 2, then your choice of stretch · 13:25–14:05

Purpose. Complete and explain the work; Phase 2 for those on it; stretch (macros, snapshots, incremental, unit tests) for fast finishers.

On screen. Anchor › Launch box and walkthrough cells; learners in challenges/v1/02-dbt-pipeline

Instructor notes.

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- About three and a half hours of practice across the day in total; nobody has to finish Phase 2.
- Early finishers investigate one planted error in a copied project and show the failure and the repair.

Say.

- “A successful build is not evidence. Show me one value you checked a different way.”

Say.

- “Carry on where you stopped. Phase 2 first; nobody has to finish it.”
- “Finished C10? Ask Copilot to add one of the pieces you just saw to your project: a unit test for stg_holdings is the quickest.”

Transition. Break, then the recap.

> ☕ Break · 15 min · back at 14:20

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4. Consolidation

Activity 9b — second look at yesterday's AI-generated dbt · 14:20–14:35

Purpose. Learners reopen the dbt project they generated yesterday, return to the two things they wrote down at 09:00 that they could not judge, and record what they catch now that they know dbt.

On screen. Anchor › Activity 9b card; learners in the project Copilot and Catalyst generated on Day 2

Instructor notes.

- This closes the loop the room opened yesterday: 'do we trust it, review all of it, or review some of it?' The answer the activity gives is an order to review in: sources, staging, joins, tests, then anything nobody asked for.
- Yesterday some projects did not finish generating. Those learners pair with a neighbour whose project did, or review the Day 2 example in courseware/day2-platform/demos/contract-to-code/eod_prices.
- Ten minutes of review in pairs, five of shareback: two or three pairs give one thing they catch now. Things still not judged go in the Q&A table.

Say.

- "Take out the two things you wrote down this morning. Can you judge them now?"
- "You do not review five hundred files. You review in graph order, and you start with the tests: do they guard the promises in your contract?"

Activity 9b — Second look at yesterday's AI-generated dbt. Pairs, 15 min; the card is in `activities/`.

Transition. From what you caught to what we learned today.

If ahead or behind. Short of time: pairs check only their two morning notes and the tests.

Debrief — what did we learn? · 14:35–14:45

Purpose. Ask the room what it learned and fill gaps from the ten topics; arrange tomorrow's contributions (bitemporal loading, deployment).

On screen. Anchor › Recap table

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Instructor notes.

- Ask ‘what did we learn?’ and fill gaps from the recap table rather than lecturing.
- Recruit two volunteers for tomorrow: one with bitemporal loading in production, one with Harness access; give them the anchor’s brief.

Say.

- “Who here has a bitemporal table in production? Would you show us tomorrow morning: the columns and one or two loading strategies, ten minutes?”

Ask at the whiteboard — what did we learn? (fill gaps from this list, do not lecture it). Ask in order, write what the room says, add what is missing.

Ask	Expect, then add
Models and conventions	Staging and integration models; CG naming (stg_, int_, fct_, agg_; on Friday src_, dom_, rpt_); the SQL behind a transformation is a SELECT dbt wraps.
Dependencies and change	ref() and source() build the graph; a source or schema change is visible in lineage; dbt docs show it.
Testing and contracts	Schema tests and singular tests are the contract’s quality rules made executable; catalog information and descriptions come from the same YAML.
Bitemporal loading	Tomorrow morning: the columns and one or two strategies, from a volunteer or guest. Questions to carry: what is business time vs system time? what happens on a correction?
Deployment context	Tomorrow afternoon: Harness and CI/CD, with screenshots for anyone without access.
Catalyst	Generation, iteration, and the repository’s data_products area where the code goes; 80 to 90% of the code from a good contract.
Airflow preview	YAML-configured blueprints are tomorrow’s topic; today’s dbt project is what they will run.

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Ask	Expect, then add
Execution and storage	dbt SQL runs on Databricks as Spark; files on S3 in Delta: columnar, with logs and manifests that give transactions; you can insert and delete, unlike plain Parquet.
Catalogue conventions	Application, domain and source-aligned catalogs; publish via the domain catalog; the Marketplace is discovery.
Quality and materialisation	Checks at project and model level; materialisation set per folder in dbt_project.yml or in the model file.

Transition. Pairs: swap a mapping and its test.

Activity 9 — test review in pairs · 14:45–14:55

Purpose. Exchange one mapping rule and its test; propose a wrong result and decide whether the test would catch it; agree one improvement.

On screen. Anchor › Activity 9 card

Instructor notes.

- The three wrong results: a duplicated join, a missing dated FX rate, a changed unit.

Say.

- “Would your test catch it? If not, write the expected failure before you write the fix.”

Activity 9 — Test review in pairs. Pairs, 10 min; the card is in `activities/`.

Transition. Two or three examples to the room, then close.

Close — closing discussion · 14:55–15:00

Purpose. Share a missing check; save the dbt work for orchestration; confirm tomorrow’s baseline.

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On screen. Anchor › Before learners leave

Instructor notes.

- Completing advanced dbt is not a prerequisite for tomorrow; the Airflow challenge ships the solved dbt project.

Say.

- “Keep this project where it is. Tomorrow Airflow runs it.”

Say.

- “Two or three examples to the room: the wrong result, whether the test caught it, the improvement.”
- “Save the project where it is; tomorrow Airflow runs it. Completing Phase 2 is not a prerequisite for tomorrow: the Airflow challenge ships the solved dbt project.”

Before learners leave

- [ ] The two things each learner could not judge in yesterday’s generated project, marked as judged or still open, and one thing they catch now.
- [ ] The working dbt project, model and test results, one independent output check and the outstanding questions for orchestration, kept where they are.
- [ ] Each learner can trace one plain-English mapping through its model and its test.
- [ ] Two volunteers (or a guest) confirmed for tomorrow’s bitemporal and deployment slots, with the brief.
- [ ] The Airflow baseline is ready: Databricks auth works, challenge 02’s seeds are loaded, and Astro has been started once after setup.

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